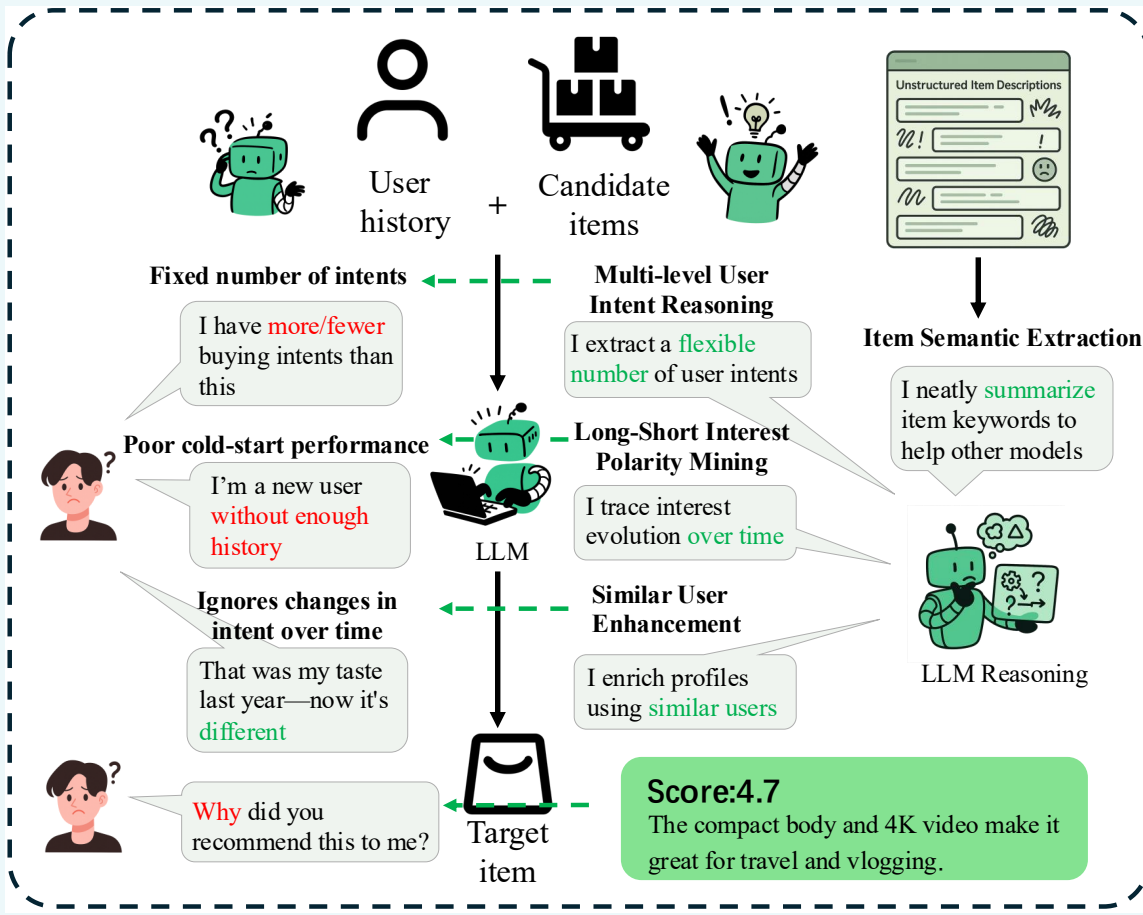


Problem and Motivation

Sequential recommendation still struggles with:

- Evidence insufficiency:** cold-start sparsity and noisy item text weaken candidate evidence.
- Opaque intent modeling:** long- and short-horizon preferences are collapsed into one latent state.
- Fragmented fixes:** prior methods address content, intent, or collaboration separately rather than within one reasoning loop.



Visual takeaway. Sparse history, shifting intent, and weak item semantics interact with each other, so R³-REC handles them in one connected evidence pipeline instead of with isolated fixes.

What changes in R³-REC.

- recover usable evidence from sparse and noisy histories;
- separate stable intent from short-term polarity shifts;
- ask the LLM to judge explicit clues rather than hidden vectors alone.

Core idea. R³-REC turns ranking into retrieval-augmented evidence reasoning: infer explicit intent, refine item semantics, retrieve similar-user evidence, and score candidates with a compact LLM prompt.

Headline Numbers

+10.2% HR@1 gain
+6.4% HR@5 gain
580 ms ranking latency

Key Contributions

- Reasoning-driven recommendation:** a prompt-centric pipeline that separates intent inference, text refinement, retrieval, and final scoring.
- Multi-granular user modeling:** macro intent weights and long/short polarity vectors capture stable and dynamic preferences simultaneously.
- Content plus collaboration:** compact item keyphrases and readable similar-user sketches strengthen sparse-history recommendation.
- Practical effectiveness:** strong gains over neural and LLM baselines while keeping inference within sub-second latency.

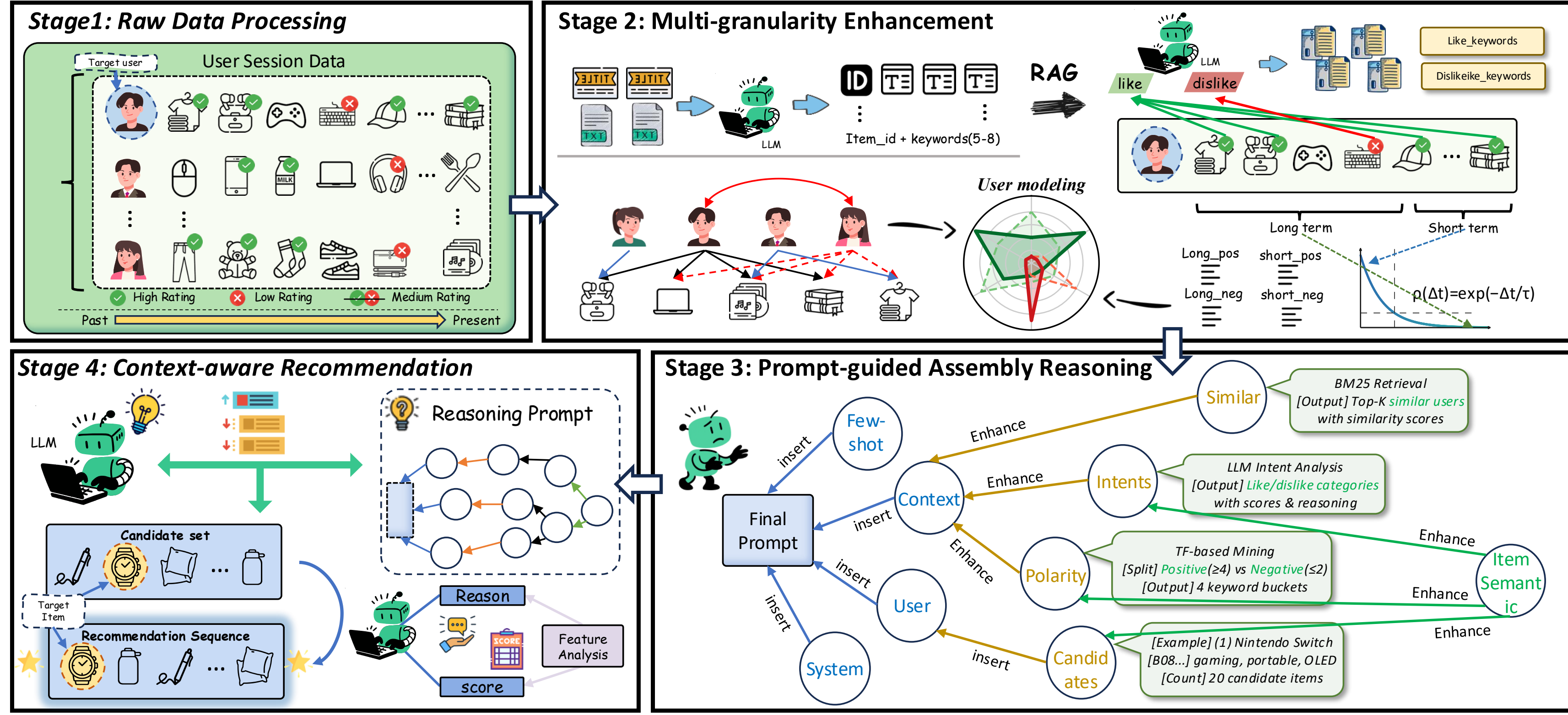
Experimental Setup and Practicality

- Datasets:** *ML-1M*, *Games*, and *Bundle*.
- Protocol:** histories truncated to $H_{\max} = 100$; each session is ranked over a fixed 20-item candidate set with 8:1:1 train/validation/test splits.
- Metrics:** HR@1, HR@5, and HR@10.
- Baselines:** conventional methods, single-/multi-intent models, and recent LLM recommenders including NIR, PO4ISR, RecMind, and UniRec.
- LLM backbone:** GPT-3.5-Turbo with deterministic inference.

Model	Tokens (I/O)	Latency	HR@1
NIR	150 / 10	250 ms	0.1168
PO4ISR	350 / 15	380 ms	0.2588
RecMind	600 / 80	920 ms	0.3450
R ³ -REC	1250 / 10	580 ms	0.3790

Retrieval-heavy prompting remains practical because prefilling is parallelizable and the model only needs to emit short scoring outputs rather than long reasoning traces.

Methodology Overview



Stage 1: Hierarchical intent reasoning. Contrastive category evidence produces a stable macro-intent representation $\mathbf{z}_u$.

Stage 2: Long-short polarity mining. Positive vs. negative keyword centroids across short and long windows form directional preference signals $\mathbf{q}_u^{(S)}$ and $\mathbf{q}_u^{(L)}$.

Stage 3: Similar-user collaborative enhancement. Hybrid dense retrieval plus BM25 recalls readable neighbor summaries instead of opaque latent vectors.

Stage 4: Reasoning-based interest matching. The LLM inspects intent clues, collaborative evidence, and item keyphrases, then emits a calibrated match score with a concise rationale.

Why the Pipeline Works

Compact Evidence Construction

- Item semantic extraction:** each item is distilled to 5–8 high-value keyphrases, reducing prompt noise while preserving coverage.
- Hybrid retrieval:** textual user sketches combine semantic similarity and exact lexical overlap for cold-start robustness.

Intent State

$$\mathbf{h}_u = \mathbf{W}_h \left[\mathbf{z}_u; \mathbf{q}_u^{(S)}; \mathbf{q}_u^{(L)} \right] + \mathbf{b}_h$$

The lightweight adapter aligns explicit intent and polarity evidence before retrieval and final prompting.

Scoring Logic

- The prompt contains explicit user intents, polarity cues, retrieved similar-user summaries, and item keyphrases.
- Coverage features quantify how well candidate attributes match the user's positive and negative preference keywords.

Deployment Benefits

- Cheap output, richer input:** complexity is shifted to retrieval and prompt assembly rather than long chain-of-thought generation.
- Interpretability:** every score is grounded in readable intents, keyphrases, and retrieved neighbor evidence.

Detailed Method Formulation

1. Intent Reasoning and Polarity

The model explicitly separates stable category-level intent from dynamic attribute-level preference shifts.

$$\pi_k = \frac{\exp(\kappa(g_k^{(+)} - g_k^{(-)}))}{\sum_{k' \in \mathcal{K}_u} \exp(\kappa(g_{k'}^{(+)} - g_{k'}^{(-)}))}$$

Category weights suppress mixed-feedback noise, then aggregate into a stable user-intent anchor:

$$\mathbf{z}_u = \sum_{k \in \mathcal{K}_u} \pi_k \mathbf{e}_k$$

For short- and long-term horizons, positive and negative keyword centroids form directional polarity vectors:

$$\mathbf{q}_u^{(\ell)} = \mathbf{p}_+^{(\ell)} - \mathbf{p}_-^{(\ell)}, \quad \ell \in \{S, L\}$$

These structured signals are serialized into one final prompt, where the LLM acts as a **judge** rather than a generator: it verifies evidence consistency, emits a categorical verdict, and maps token probabilities to the final recommendation score.

2. Item, Retrieval, and Matching

Item descriptions are compressed into non-redundant high-value keyphrases by maximizing relevance and coverage:

$$\max_{S_i \subseteq \mathcal{U}} \left(\sum_{c \in S_i} \psi(c) + \lambda \sum_{q \in \mathcal{U}_i} \max_{c \in S_i} \cos(\mathbf{v}_q, \mathbf{v}_c) \right)$$

Similar users are then recalled with a hybrid dense-plus-sparse retrieval score:

$$\text{sim}(u, v) = \lambda \frac{\text{BM25}(\mathbf{s}_u, \mathbf{s}_v)}{Z_{\text{bm}}} + (1 - \lambda) \cos(\mathbf{m}_u, \mathbf{m}_v)$$

Before scoring, explicit coverage features measure how strongly candidate attributes align with positive or negative user evidence:

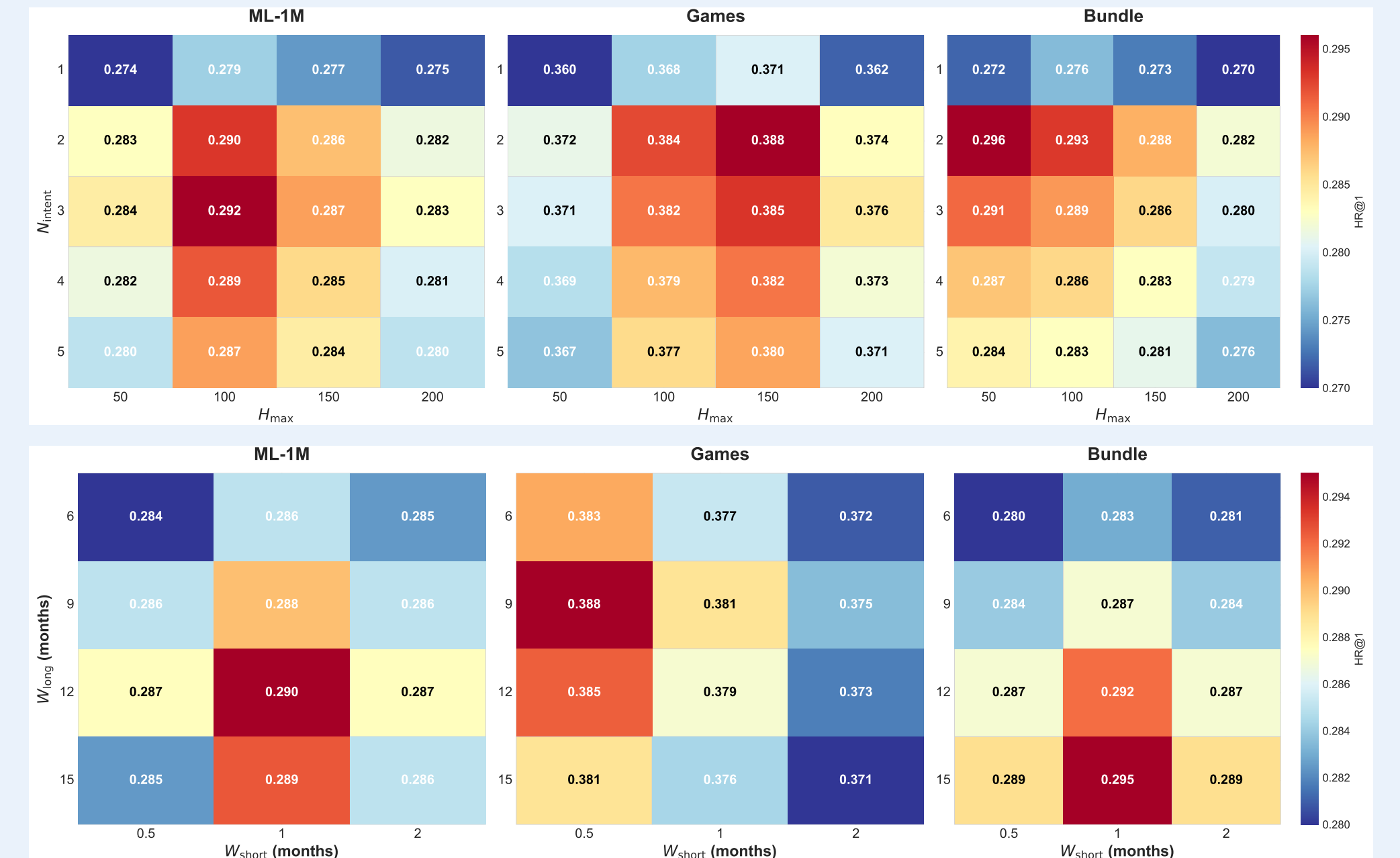
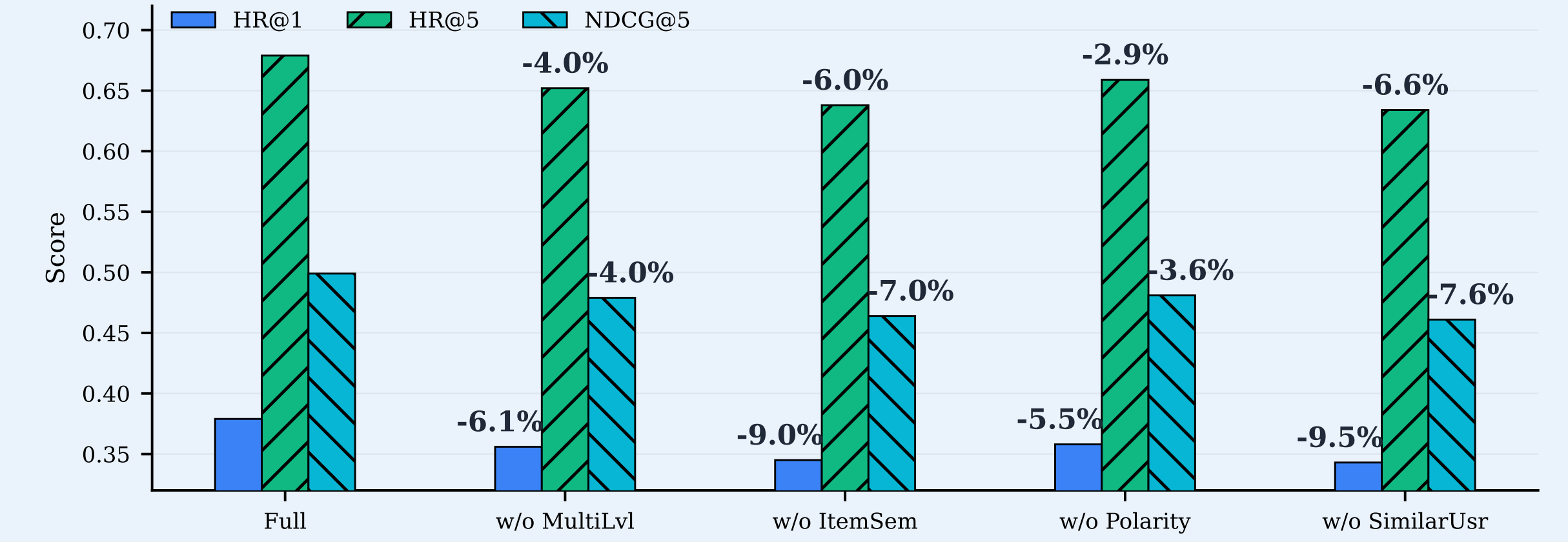
$$\text{cov}^{\pm}(u, i) = \max_{w \in \mathcal{K}_u^{\pm}} \max_{k \in S_i} \cos(\mathbf{e}_w, \mathbf{e}_k)$$

Main Comparison Table

		Conventional		Single-Intent		Multi-Intent			LLM Methods				Ours	Gain	
Data	Metric	MostPop	FPMC	NARM	STAMP	GCE-GNN	MCPRN	HIDE	Atten-Mixer	NIR	PO4ISR	RecMind	UniRec	R ³ -REC	
ML-1M	HR@1	0.0004	0.1132	0.1692*	0.1584	0.1312	0.1434	0.1498	0.1490	0.0572	0.2000	0.2700 [†]	0.2650	0.287	+6.3%
	HR@5	0.0070	0.3748	0.5230*	0.5078	0.4748	0.4788	0.4998	0.4932	0.2326	0.5510	0.5720	0.5910 [†]	0.597	+1.0%
	HR@10	0.0120	0.5150	0.7050*	0.6920	0.6600	0.6640	0.6820	0.6760	0.3550	0.7240	0.7480	0.7620 [†]	0.769	+0.9%
Games	HR@1	–	0.0498	0.0572	0.0556	0.0692	0.0522	0.0696	0.0530	0.1168*	0.2588	0.3450 [†]	0.3400	0.379	+9.9%
	HR@5	–	0.2564	0.2574	0.2586	0.2744	0.2416	0.2694	0.2472	0.3406*	0.5866	0.6550 [†]	0.6450	0.679	+3.7%
	HR@10	–	0.4020	0.4050	0.4060	0.4260	0.3900	0.4180	0.3980	0.5100	0.7600	0.7920 [†]	0.7800	0.812	+2.5%
Bundle	HR@1	–	0.0398	0.0322	0.0365	0.0360	0.0360	0.0458	0.0525	0.0975*	0.1697	0.2550 [†]	0.2450	0.281	+10.2%
	HR@5	0.0042	0.2475	0.2332	0.2352	0.2237	0.2352	0.2585	0.2644	0.2832*	0.4328	0.5150 [†]	0.5050	0.548	+6.4%
	HR@10	0.0070	0.3820	0.3650	0.3680	0.3500	0.3680	0.3940	0.4020	0.4300	0.6120	0.6650 [†]	0.6550	0.703	+5.7%

Best results are in bold. Gains are strongest on sparse and noisy domains, showing that explicit reasoning over refined evidence is particularly effective when user history is limited.

Ablation and Robustness



- Removing **item semantic extraction** or **similar-user enhancement** causes the largest performance drops, confirming that both refined content cues and collaborative evidence are essential.
- Best intent count, history depth, and temporal windows are dataset-specific, but the model remains stable: moving away from the optimum changes HR@1 by at most about 1.8 percentage points.
- The recommended operating region is compact: **5–8 keyphrases** per item and a moderate BM25 cut-off already saturate performance.

Result Interpretation and Robustness Summary

Performance reading. The largest improvements appear on *Games* and *Bundle*, especially on HR@1, indicating that retrieval-augmented explicit evidence is most helpful when histories are sparse or noisy. R³-REC also outperforms RecMind while remaining substantially faster, because reasoning is pushed into evidence preparation rather than long output generation.

Robustness reading. Removing **item semantic extraction** or **similar-user enhancement** causes the largest drop, confirming that refined content cues and collaborative retrieval are both essential. Hyper-parameters are dataset-specific, but stability is good: moving away from the optimum changes HR@1 by at most about **1.8 percentage points**, and **5–8 keyphrases** per item already provide a strong coverage-budget trade-off.

Future outlook. A natural next step is to reuse the same explicit intent and evidence construction pipeline for a lighter non-LLM ranker or online retriever, improving scalability while preserving interpretability.